

Barcode Crossover Analysis

Method:

I aligned the reads from before and after the protocol tweaks to each of the reference sequences. Then I ran alignment analysis on each to ascertain the number of mapped reads to each reference. Lastly, I calculated the total crossover percent as the number of reads mapped to the non-D5405 reference sequence divided by the total number of reads mapped.

Analysis:

Mapped Reads		
Sample	1st Run	2nd Run
Human	205	18
Rice	109	39
Lambda	40	7
D5405	91656	98794
Total	92010	98858
Crossover Percent		
Total	0.385	0.065

The total crossover from the 1st run was 0.385% while the total crossover from the 2nd run was 0.065%.

Conclusion:

The tweaks made to the protocol have greatly reduced the occurrence of barcode crossover.